

Lexical Grammar

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1 Regex

1.1 Substitutions

L	$:=$	$[a - z]$	(1)
U	$:=$	$[A - Z]$	(2)
D	$:=$	$[0 - 9]$	(3)
H	$:=$	$[a - fA - F0 - 9]$	(4)
O	$:=$	$[0 - 7]$	(5)
B	$:=$	$[01]$	(6)

1.2 Identifiers, Constants, and Comments

$_{-}\{0, 2\}L(? : L U D _{-})^{*}$	$\rightarrow$	IDENT	(7)
$U(? : L U D)^{*}$	$\rightarrow$	TYPE_IDENTIFIER	(8)
$[\backslash + -]?D+$	$\rightarrow$	NUMERIC_CONST Integer	(9)
$0[xX]H\{1, 16\}$	$\rightarrow$	NUMERIC_CONST Hex	(10)
$0[oO]O\{1, 32\}$	$\rightarrow$	NUMERIC_CONST Octal	(11)
$0[bB]B\{1, 64\}$	$\rightarrow$	NUMERIC_CONST Binary	(12)
$[\backslash + -]?D + [eE][\backslash + -]?D+$	$\rightarrow$	NUMERIC_CONST Scientific Notation	(13)
$[\backslash + -]?(? : D + (? : [.]D^{*})?[.]D+)$	$\rightarrow$	NUMERIC_CONST Float	(14)
$r?"((? : [^ " \backslash] \backslash \backslash ")^{*})"$	$\rightarrow$	STRING_LITERAL	(15)
$\#. * \backslash n\$$	$\rightarrow$	COMMENT	(16)

1.3 Symbols

,	$\rightarrow$	COMMA	(17)
;	$\rightarrow$	SEMI	(18)
(	$\rightarrow$	LEFT_PAREN	(19)
)	$\rightarrow$	RIGHT_PAREN	(20)
[	$\rightarrow$	LEFT_SQUARE	(21)
]	$\rightarrow$	RIGHT_SQUARE	(22)
{	$\rightarrow$	LEFT_CURLY	(23)
}	$\rightarrow$	RIGHT_CURLY	(24)

1.4 Keywords

and	→	LOGICAL_AND	(25)
bool	→	BOOL	(26)
break	→	BREAK	(27)
class	→	CLASS	(28)
elif	→	ELIF	(29)
else	→	ELSE	(30)
enum	→	ENUM	(31)
exit	→	EXIT	(32)
false	→	FALSE	(33)
float32	→	FLOAT32	(34)
float64	→	FLOAT64	(35)
for	→	FOR	(36)
func	→	FUNCTION	(37)
if	→	IF	(38)
iface	→	INTERFACE	(39)
in	→	IN	(40)
int16	→	INT16	(41)
int32	→	INT32	(42)
int64	→	INT64	(43)
int8	→	INT8	(44)
mut	→	MUTABLE	(45)
or	→	LOGICAL_OR	(46)
private	→	PRIVATE	(47)
public	→	PUBLIC	(48)
string	→	STRING	(49)
true	→	TRUE	(50)
uint16	→	UINT16	(51)
uint32	→	UINT32	(52)
uint64	→	UINT64	(53)
uint8	→	UINT8	(54)
while	→	WHILE	(55)

1.5 Operators

$\wedge\wedge=$	→	EXPONENT_ASSIGN	(56)
$\wedge\wedge$	→	EXPONENT	(57)
$\wedge=$	→	XOR_ASSIGN	(58)
$\wedge$	→	XOR	(59)
$\% =$	→	MOD_ASSIGN	(60)
$\%$	→	MOD	(61)
$* =$	→	MUL_ASSIGN	(62)
$*$	→	MUL	(63)
$// =$	→	INTEGER_DIV_ASSIGN	(64)
$//$	→	INTEGER_DIV	(65)
$/ =$	→	DIV_ASSIGN	(66)
$/$	→	DIV	(67)
$++$	→	INC	(68)
$+ =$	→	ADD_ASSIGN	(69)
$+$	→	ADD	(70)
$--$	→	DEC	(71)
$- =$	→	SUB_ASSIGN	(72)
$-$	→	SUB	(73)
$<<$	→	SHIFT_LEFT	(74)
$<=$	→	LE_OP	(75)
$<$	→	LT_OP	(76)
$>>$	→	SHIFT_RIGHT	(77)
$>=$	→	GE_OP	(78)
$>$	→	GT_OP	(79)
$!=$	→	NE_OP	(80)
$==$	→	EQ_OP	(81)
$=$	→	ASSIGN	(82)
$\& =$	→	AND_ASSIGN	(83)
$\&$	→	AND	(84)
$ =$	→	OR_ASSIGN	(85)
$ $	→	OR	(86)
$?$	→	TERNARY	(87)
$.$	→	DOT	(88)

2 Backus-Naur Form

$\langle \text{ident} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(89)
		$ \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(90)
		$ \text{"_"} \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(91)
$\langle \text{ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(92)
		$ \langle \text{upper} \rangle \langle \text{ident?} \rangle$	(93)
		$ \langle \text{digit} \rangle \langle \text{ident?} \rangle$	(94)
		$ \text{"_"} \langle \text{ident?} \rangle$	(95)
		$ \varepsilon$	(96)
$\langle \text{type_ident} \rangle$	$::=$	$\langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(97)
$\langle \text{type_ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{type_ident?} \rangle$	(98)
		$ \langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(99)
		$ \langle \text{digit} \rangle \langle \text{type_ident?} \rangle$	(100)
		$ \varepsilon$	(101)
$\langle \text{float_const} \rangle$	$::=$	$\langle \text{mantissa} \rangle$	(102)
		$ \langle \text{hex_const} \rangle$	(103)
		$ \langle \text{octal_const} \rangle$	(104)
		$ \langle \text{binary_const} \rangle$	(105)
		$ \langle \text{scientific} \rangle$	(106)
$\langle \text{integer_const} \rangle$	$::=$	$\langle \text{decimal_const} \rangle$	(107)
		$ \langle \text{hex_const} \rangle$	(108)
		$ \langle \text{octal_const} \rangle$	(109)
		$ \langle \text{binary_const} \rangle$	(110)
$\langle \text{comment} \rangle$	$::=$	$\text{"\#"} \langle \text{comment_text} \rangle$	(111)
$\langle \text{comment_text} \rangle$	$::=$	$\langle \text{comment_char} \rangle \langle \text{comment_text} \rangle$	(112)
$\langle \text{comment_char} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{upper} \rangle \langle \text{digit} \rangle \langle \text{symbol} \rangle \langle \text{non_newline} \rangle$	(113)
$\langle \text{string_literal} \rangle$	$::=$	$\langle \text{r?} \rangle \langle \text{quote} \rangle \langle \text{content} \rangle \langle \text{quote} \rangle$	(114)
$\langle \text{content} \rangle$	$::=$	$\langle \text{char} \rangle \langle \text{content} \rangle \varepsilon$	(115)
$\langle \text{char} \rangle$	$::=$	$\langle \text{escaped_quote} \rangle \langle \text{normal_symbol} \rangle$	(116)
$\langle \text{scientific} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{mantissa} \rangle \langle \text{exponent} \rangle$	(117)
$\langle \text{mantissa} \rangle$	$::=$	$\langle \text{digits} \rangle \text{"."} \langle \text{digits?} \rangle$	(118)
		$ \text{"."} \langle \text{digits} \rangle$	(119)
		$ \langle \text{digits} \rangle$	(120)
$\langle \text{exponent} \rangle$	$::=$	$\text{"e"} \langle \text{decimal_const} \rangle$	(121)
$\langle \text{decimal_const} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{digits} \rangle$	(122)
$\langle \text{hex_const} \rangle$	$::=$	$\text{"0"} \text{"x"} \langle \text{hs} \rangle$	(123)
		$ \text{"0"} \text{"X"} \langle \text{hs} \rangle$	(124)
$\langle \text{octal_const} \rangle$	$::=$	$\text{"0"} \text{"o"} \langle \text{os} \rangle$	(125)
		$ \text{"0"} \text{"O"} \langle \text{os} \rangle$	(126)
$\langle \text{binary_const} \rangle$	$::=$	$\text{"0"} \text{"b"} \langle \text{bs} \rangle$	(127)
		$ \text{"0"} \text{"B"} \langle \text{bs} \rangle$	(128)

<code>< and ></code>	<code>::=</code>	<code>"a" "n" "d"</code>	(129)
<code>< bool ></code>	<code>::=</code>	<code>"b" "o" "o" "l"</code>	(130)
<code>< break ></code>	<code>::=</code>	<code>"b" "r" "e" "a" "k"</code>	(131)
<code>< class ></code>	<code>::=</code>	<code>"c" "l" "a" "s" "s"</code>	(132)
<code>< elif ></code>	<code>::=</code>	<code>"e" "l" "i" "f"</code>	(133)
<code>< else ></code>	<code>::=</code>	<code>"e" "l" "s" "e"</code>	(134)
<code>< enum ></code>	<code>::=</code>	<code>"e" "n" "u" "m"</code>	(135)
<code>< exit ></code>	<code>::=</code>	<code>"e" "x" "i" "t"</code>	(136)
<code>< false ></code>	<code>::=</code>	<code>"f" "a" "l" "s" "e"</code>	(137)
<code>< float32 ></code>	<code>::=</code>	<code>"f" "l" "o" "a" "t" "3" "2"</code>	(138)
<code>< float64 ></code>	<code>::=</code>	<code>"f" "l" "o" "a" "t" "6" "4"</code>	(139)
<code>< for ></code>	<code>::=</code>	<code>"f" "o" "r"</code>	(140)
<code>< func ></code>	<code>::=</code>	<code>"f" "u" "n" "c"</code>	(141)
<code>< if ></code>	<code>::=</code>	<code>"i" "f"</code>	(142)
<code>< iface ></code>	<code>::=</code>	<code>"i" "f" "a" "c" "e"</code>	(143)
<code>< in ></code>	<code>::=</code>	<code>"i" "n"</code>	(144)
<code>< int16 ></code>	<code>::=</code>	<code>"i" "n" "t" "1" "6"</code>	(145)
<code>< int32 ></code>	<code>::=</code>	<code>"i" "n" "t" "3" "2"</code>	(146)
<code>< int64 ></code>	<code>::=</code>	<code>"i" "n" "t" "6" "4"</code>	(147)
<code>< int8 ></code>	<code>::=</code>	<code>"i" "n" "t" "8"</code>	(148)
<code>< mut ></code>	<code>::=</code>	<code>"m" "u" "t"</code>	(149)
<code>< or ></code>	<code>::=</code>	<code>"o" "r"</code>	(150)
<code>< private ></code>	<code>::=</code>	<code>"p" "r" "i" "v" "a" "t" "e"</code>	(151)
<code>< public ></code>	<code>::=</code>	<code>"p" "u" "b" "l" "i" "c"</code>	(152)
<code>< string ></code>	<code>::=</code>	<code>"s" "t" "r" "i" "n" "g"</code>	(153)
<code>< true ></code>	<code>::=</code>	<code>"t" "r" "u" "e"</code>	(154)
<code>< uint16 ></code>	<code>::=</code>	<code>"u" "i" "n" "t" "1" "6"</code>	(155)
<code>< uint32 ></code>	<code>::=</code>	<code>"u" "i" "n" "t" "3" "2"</code>	(156)
<code>< uint64 ></code>	<code>::=</code>	<code>"u" "i" "n" "t" "6" "4"</code>	(157)
<code>< uint8 ></code>	<code>::=</code>	<code>"u" "i" "n" "t" "8"</code>	(158)
<code>< while ></code>	<code>::=</code>	<code>"w" "h" "i" "l" "e"</code>	(159)

$\langle \text{exponent_assign} \rangle$	::=	" ^ " " ^ " " = "	(160)
$\langle \text{exponent} \rangle$	::=	" ^ " " ^ "	(161)
$\langle \text{xor_assign} \rangle$	::=	" ^ " " = "	(162)
$\langle \text{xor} \rangle$	::=	" ^ "	(163)
$\langle \text{mod_assign} \rangle$	::=	" % " " = "	(164)
$\langle \text{mod} \rangle$	::=	" % "	(165)
$\langle \text{mul_assign} \rangle$	::=	" * " " = "	(166)
$\langle \text{mul} \rangle$	::=	" * "	(167)
$\langle \text{integer_div_assign} \rangle$	::=	" / " " / " " = "	(168)
$\langle \text{integer_div} \rangle$	::=	" / " " / "	(169)
$\langle \text{div_assign} \rangle$	::=	" / " " = "	(170)
$\langle \text{div} \rangle$	::=	" / "	(171)
$\langle \text{inc} \rangle$	::=	" + " " + "	(172)
$\langle \text{add_assign} \rangle$	::=	" + " " = "	(173)
$\langle \text{add} \rangle$	::=	" + "	(174)
$\langle \text{dec} \rangle$	::=	" - " " - "	(175)
$\langle \text{sub_assign} \rangle$	::=	" - " " = "	(176)
$\langle \text{sub} \rangle$	::=	" - "	(177)
$\langle \text{shift_left} \rangle$	::=	" < " " < "	(178)
$\langle \text{le_op} \rangle$	::=	" < " " = "	(179)
$\langle \text{lt_op} \rangle$	::=	" < "	(180)
$\langle \text{shift_right} \rangle$	::=	" > " " > "	(181)
$\langle \text{ge_op} \rangle$	::=	" > " " = "	(182)
$\langle \text{gt_op} \rangle$	::=	" > "	(183)
$\langle \text{ne_op} \rangle$	::=	" ! " " = "	(184)
$\langle \text{eq_op} \rangle$	::=	" = " " = "	(185)
$\langle \text{assign} \rangle$	::=	" = "	(186)
$\langle \text{and_assign} \rangle$	::=	" & " " = "	(187)
$\langle \text{and} \rangle$	::=	" & "	(188)
$\langle \text{or_assign} \rangle$	::=	" " " = "	(189)
$\langle \text{or} \rangle$	::=	" "	(190)
$\langle \text{ternary} \rangle$	::=	" ? "	(191)
$\langle \text{dot} \rangle$	::=	" . "	(192)

$\langle \text{digits?} \rangle$	$::= \langle \text{digits} \rangle \mid \varepsilon$	(193)
$\langle \text{digits} \rangle$	$::= \langle \text{digit} \rangle \mid \langle \text{digit} \rangle \langle \text{digits} \rangle$	(194)
$\langle \text{sign?} \rangle$	$::= \langle \text{sign} \rangle \mid \varepsilon$	(195)
$\langle \text{sign} \rangle$	$::= "+" \mid "-"$	(196)
$\langle \text{lower} \rangle$	$::= "a" \mid "b" \mid "c" \mid "d" \mid "e" \mid "f" \mid "g" \mid "h" \mid "i" \mid "j" \mid "k" \mid "l" \mid "m" \mid$ $"n" \mid "o" \mid "p" \mid "q" \mid "r" \mid "s" \mid "t" \mid "u" \mid "v" \mid "w" \mid "x" \mid "y" \mid "z"$	(197)
$\langle \text{upper} \rangle$	$::= "A" \mid "B" \mid "C" \mid "D" \mid "E" \mid "F" \mid "G" \mid "H" \mid "I" \mid "J" \mid "K" \mid "L" \mid "M" \mid$ $"N" \mid "O" \mid "P" \mid "Q" \mid "R" \mid "S" \mid "T" \mid "U" \mid "V" \mid "W" \mid "X" \mid "Y" \mid "Z"$	(198)
$\langle \text{digit} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7" \mid "8" \mid "9"$	(199)
$\langle \text{hex} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7" \mid "8" \mid "9" \mid$ $"a" \mid "b" \mid "c" \mid "d" \mid "e" \mid "f" \mid "A" \mid "B" \mid "C" \mid "D" \mid "E" \mid "F"$	(200)
$\langle \text{octal} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7"$	(201)
$\langle \text{binary} \rangle$	$::= "0" \mid "1"$	(202)
$\langle \text{symbols} \rangle$	$::= \langle \text{normal_symbol} \rangle \mid \langle \text{quote} \rangle \mid \langle \text{backslash} \rangle$	(203)
$\langle \text{normal_symbol} \rangle$	$::= "!" \mid "\#" \mid "$" \mid "\%" \mid "\&" \mid "'" \mid "(" \mid ")" \mid "*" \mid "+" \mid "," \mid "-" \mid "." \mid "/" \mid ":" \mid ";" \mid$ $"<" \mid "=" \mid ">" \mid "?" \mid "@" \mid "[" \mid "]" \mid "[" \mid "]" \mid "_" \mid "`" \mid "{" \mid "}" \mid " " \mid "$	(204)
$\langle \text{whitespace} \rangle$	$::= \langle \text{non_newline} \rangle \mid \langle \text{newline} \rangle$	(205)
$\langle \text{non_newline} \rangle$	$::= " " \mid "\backslash t" \mid "\backslash r"$	(206)
$\langle \text{newline} \rangle$	$::= "\backslash n"$	(207)
$\langle \text{r?} \rangle$	$::= "r" \mid \varepsilon$	(208)
$\langle \text{escaped_quote} \rangle$	$::= \langle \text{backslash} \rangle \langle \text{quote} \rangle$	(209)
$\langle \text{backslash} \rangle$	$::= "\backslash"$	(210)
$\langle \text{quote} \rangle$	$::= "\""$	(211)
		(212)

$\langle hs \rangle$	$::=$	$\langle hex \rangle$	(213)
		$ \langle hex \rangle \langle hs1 \rangle$	(214)
$\langle hs1 \rangle$	$::=$	$\langle hex \rangle$	(215)
		$ \langle hex \rangle \langle hs2 \rangle$	(216)
$\langle hs2 \rangle$	$::=$	$\langle hex \rangle$	(217)
		$ \langle hex \rangle \langle hs3 \rangle$	(218)
$\langle hs3 \rangle$	$::=$	$\langle hex \rangle$	(219)
		$ \langle hex \rangle \langle hs4 \rangle$	(220)
$\langle hs4 \rangle$	$::=$	$\langle hex \rangle$	(221)
		$ \langle hex \rangle \langle hs5 \rangle$	(222)
$\langle hs5 \rangle$	$::=$	$\langle hex \rangle$	(223)
		$ \langle hex \rangle \langle hs6 \rangle$	(224)
$\langle hs6 \rangle$	$::=$	$\langle hex \rangle$	(225)
		$ \langle hex \rangle \langle hs7 \rangle$	(226)
$\langle hs7 \rangle$	$::=$	$\langle hex \rangle$	(227)
		$ \langle hex \rangle \langle hs8 \rangle$	(228)
$\langle hs8 \rangle$	$::=$	$\langle hex \rangle$	(229)
		$ \langle hex \rangle \langle hs9 \rangle$	(230)
$\langle hs9 \rangle$	$::=$	$\langle hex \rangle$	(231)
		$ \langle hex \rangle \langle hs10 \rangle$	(232)
$\langle hs10 \rangle$	$::=$	$\langle hex \rangle$	(233)
		$ \langle hex \rangle \langle hs11 \rangle$	(234)
$\langle hs11 \rangle$	$::=$	$\langle hex \rangle$	(235)
		$ \langle hex \rangle \langle hs12 \rangle$	(236)
$\langle hs12 \rangle$	$::=$	$\langle hex \rangle$	(237)
		$ \langle hex \rangle \langle hs13 \rangle$	(238)
$\langle hs13 \rangle$	$::=$	$\langle hex \rangle$	(239)
		$ \langle hex \rangle \langle hs14 \rangle$	(240)
$\langle hs14 \rangle$	$::=$	$\langle hex \rangle$	(241)
		$ \langle hex \rangle \langle hs15 \rangle$	(242)
$\langle hs15 \rangle$	$::=$	$\langle hex \rangle$	(243)

$\langle os \rangle$	$::=$	$\langle octal \rangle$	(244)
		$ \langle octal \rangle \langle os1 \rangle$	(245)
$\langle os1 \rangle$	$::=$	$\langle octal \rangle$	(246)
		$ \langle octal \rangle \langle os2 \rangle$	(247)
$\langle os2 \rangle$	$::=$	$\langle octal \rangle$	(248)
		$ \langle octal \rangle \langle os3 \rangle$	(249)
$\langle os3 \rangle$	$::=$	$\langle octal \rangle$	(250)
		$ \langle octal \rangle \langle os4 \rangle$	(251)
$\langle os4 \rangle$	$::=$	$\langle octal \rangle$	(252)
		$ \langle octal \rangle \langle os5 \rangle$	(253)
$\langle os5 \rangle$	$::=$	$\langle octal \rangle$	(254)
		$ \langle octal \rangle \langle os6 \rangle$	(255)
$\langle os6 \rangle$	$::=$	$\langle octal \rangle$	(256)
		$ \langle octal \rangle \langle os7 \rangle$	(257)
$\langle os7 \rangle$	$::=$	$\langle octal \rangle$	(258)
		$ \langle octal \rangle \langle os8 \rangle$	(259)
$\langle os8 \rangle$	$::=$	$\langle octal \rangle$	(260)
		$ \langle octal \rangle \langle os9 \rangle$	(261)
$\langle os9 \rangle$	$::=$	$\langle octal \rangle$	(262)
		$ \langle octal \rangle \langle os10 \rangle$	(263)
$\langle os10 \rangle$	$::=$	$\langle octal \rangle$	(264)
		$ \langle octal \rangle \langle os11 \rangle$	(265)
$\langle os11 \rangle$	$::=$	$\langle octal \rangle$	(266)
		$ \langle octal \rangle \langle os12 \rangle$	(267)
$\langle os12 \rangle$	$::=$	$\langle octal \rangle$	(268)
		$ \langle octal \rangle \langle os13 \rangle$	(269)
$\langle os13 \rangle$	$::=$	$\langle octal \rangle$	(270)
		$ \langle octal \rangle \langle os14 \rangle$	(271)
$\langle os14 \rangle$	$::=$	$\langle octal \rangle$	(272)
		$ \langle octal \rangle \langle os15 \rangle$	(273)
$\langle os15 \rangle$	$::=$	$\langle octal \rangle$	(274)
		$ \langle octal \rangle \langle os16 \rangle$	(275)
$\langle os16 \rangle$	$::=$	$\langle octal \rangle$	(276)
		$ \langle octal \rangle \langle os17 \rangle$	(277)
$\langle os17 \rangle$	$::=$	$\langle octal \rangle$	(278)
		$ \langle octal \rangle \langle os18 \rangle$	(279)
$\langle os18 \rangle$	$::=$	$\langle octal \rangle$	(280)
		$ \langle octal \rangle \langle os19 \rangle$	(281)
$\langle os19 \rangle$	$::=$	$\langle octal \rangle$	(282)
		$ \langle octal \rangle \langle os20 \rangle$	(283)

$\langle os20 \rangle$	$::=$	$\langle octal \rangle$	(284)
		$ \langle octal \rangle \langle os21 \rangle$	(285)
$\langle os21 \rangle$	$::=$	$\langle octal \rangle$	(286)
		$ \langle octal \rangle \langle os22 \rangle$	(287)
$\langle os22 \rangle$	$::=$	$\langle octal \rangle$	(288)
		$ \langle octal \rangle \langle os23 \rangle$	(289)
$\langle os23 \rangle$	$::=$	$\langle octal \rangle$	(290)
		$ \langle octal \rangle \langle os24 \rangle$	(291)
$\langle os24 \rangle$	$::=$	$\langle octal \rangle$	(292)
		$ \langle octal \rangle \langle os25 \rangle$	(293)
$\langle os25 \rangle$	$::=$	$\langle octal \rangle$	(294)
		$ \langle octal \rangle \langle os26 \rangle$	(295)
$\langle os26 \rangle$	$::=$	$\langle octal \rangle$	(296)
		$ \langle octal \rangle \langle os27 \rangle$	(297)
$\langle os27 \rangle$	$::=$	$\langle octal \rangle$	(298)
		$ \langle octal \rangle \langle os28 \rangle$	(299)
$\langle os28 \rangle$	$::=$	$\langle octal \rangle$	(300)
		$ \langle octal \rangle \langle os29 \rangle$	(301)
$\langle os29 \rangle$	$::=$	$\langle octal \rangle$	(302)
		$ \langle octal \rangle \langle os30 \rangle$	(303)
$\langle os30 \rangle$	$::=$	$\langle octal \rangle$	(304)
		$ \langle octal \rangle \langle os31 \rangle$	(305)
$\langle os31 \rangle$	$::=$	$\langle octal \rangle$	(306)
$\langle bs \rangle$	$::=$	$\langle binary \rangle$	(307)
		$ \langle binary \rangle \langle bs1 \rangle$	(308)
$\langle bs1 \rangle$	$::=$	$\langle binary \rangle$	(309)
		$ \langle binary \rangle \langle bs2 \rangle$	(310)
$\langle bs2 \rangle$	$::=$	$\langle binary \rangle$	(311)
		$ \langle binary \rangle \langle bs3 \rangle$	(312)
$\langle bs3 \rangle$	$::=$	$\langle binary \rangle$	(313)
		$ \langle binary \rangle \langle bs4 \rangle$	(314)
$\langle bs4 \rangle$	$::=$	$\langle binary \rangle$	(315)
		$ \langle binary \rangle \langle bs5 \rangle$	(316)
$\langle bs5 \rangle$	$::=$	$\langle binary \rangle$	(317)
		$ \langle binary \rangle \langle bs6 \rangle$	(318)
$\langle bs6 \rangle$	$::=$	$\langle binary \rangle$	(319)
		$ \langle binary \rangle \langle bs7 \rangle$	(320)

$\langle bs7 \rangle$	::=	$\langle binary \rangle$	(321)
		$ \langle binary \rangle \langle bs8 \rangle$	(322)
$\langle bs8 \rangle$	::=	$\langle binary \rangle$	(323)
		$ \langle binary \rangle \langle bs9 \rangle$	(324)
$\langle bs9 \rangle$	::=	$\langle binary \rangle$	(325)
		$ \langle binary \rangle \langle bs10 \rangle$	(326)
$\langle bs10 \rangle$	::=	$\langle binary \rangle$	(327)
		$ \langle binary \rangle \langle bs11 \rangle$	(328)
$\langle bs11 \rangle$	::=	$\langle binary \rangle$	(329)
		$ \langle binary \rangle \langle bs12 \rangle$	(330)
$\langle bs12 \rangle$	::=	$\langle binary \rangle$	(331)
		$ \langle binary \rangle \langle bs13 \rangle$	(332)
$\langle bs13 \rangle$	::=	$\langle binary \rangle$	(333)
		$ \langle binary \rangle \langle bs14 \rangle$	(334)
$\langle bs14 \rangle$	::=	$\langle binary \rangle$	(335)
		$ \langle binary \rangle \langle bs15 \rangle$	(336)
$\langle bs15 \rangle$	::=	$\langle binary \rangle$	(337)
		$ \langle binary \rangle \langle bs16 \rangle$	(338)
$\langle bs16 \rangle$	::=	$\langle binary \rangle$	(339)
		$ \langle binary \rangle \langle bs17 \rangle$	(340)
$\langle bs17 \rangle$	::=	$\langle binary \rangle$	(341)
		$ \langle binary \rangle \langle bs18 \rangle$	(342)
$\langle bs18 \rangle$	::=	$\langle binary \rangle$	(343)
		$ \langle binary \rangle \langle bs19 \rangle$	(344)
$\langle bs19 \rangle$	::=	$\langle binary \rangle$	(345)
		$ \langle binary \rangle \langle bs20 \rangle$	(346)
$\langle bs20 \rangle$	::=	$\langle binary \rangle$	(347)
		$ \langle binary \rangle \langle bs21 \rangle$	(348)
$\langle bs21 \rangle$	::=	$\langle binary \rangle$	(349)
		$ \langle binary \rangle \langle bs22 \rangle$	(350)
$\langle bs22 \rangle$	::=	$\langle binary \rangle$	(351)
		$ \langle binary \rangle \langle bs23 \rangle$	(352)
$\langle bs23 \rangle$	::=	$\langle binary \rangle$	(353)
		$ \langle binary \rangle \langle bs24 \rangle$	(354)
$\langle bs24 \rangle$	::=	$\langle binary \rangle$	(355)
		$ \langle binary \rangle \langle bs25 \rangle$	(356)
$\langle bs25 \rangle$	::=	$\langle binary \rangle$	(357)
		$ \langle binary \rangle \langle bs26 \rangle$	(358)
$\langle bs26 \rangle$	::=	$\langle binary \rangle$	(359)
		$ \langle binary \rangle \langle bs27 \rangle$	(360)

$\langle bs27 \rangle$	::=	$\langle binary \rangle$	(361)
		$ \langle binary \rangle \langle bs28 \rangle$	(362)
$\langle bs28 \rangle$	::=	$\langle binary \rangle$	(363)
		$ \langle binary \rangle \langle bs29 \rangle$	(364)
$\langle bs29 \rangle$	::=	$\langle binary \rangle$	(365)
		$ \langle binary \rangle \langle bs30 \rangle$	(366)
$\langle bs30 \rangle$	::=	$\langle binary \rangle$	(367)
		$ \langle binary \rangle \langle bs31 \rangle$	(368)
$\langle bs31 \rangle$	::=	$\langle binary \rangle$	(369)
		$ \langle binary \rangle \langle bs32 \rangle$	(370)
$\langle bs32 \rangle$	::=	$\langle binary \rangle$	(371)
		$ \langle binary \rangle \langle bs33 \rangle$	(372)
$\langle bs33 \rangle$	::=	$\langle binary \rangle$	(373)
		$ \langle binary \rangle \langle bs34 \rangle$	(374)
$\langle bs34 \rangle$	::=	$\langle binary \rangle$	(375)
		$ \langle binary \rangle \langle bs35 \rangle$	(376)
$\langle bs35 \rangle$	::=	$\langle binary \rangle$	(377)
		$ \langle binary \rangle \langle bs36 \rangle$	(378)
$\langle bs36 \rangle$	::=	$\langle binary \rangle$	(379)
		$ \langle binary \rangle \langle bs37 \rangle$	(380)
$\langle bs37 \rangle$	::=	$\langle binary \rangle$	(381)
		$ \langle binary \rangle \langle bs38 \rangle$	(382)
$\langle bs38 \rangle$	::=	$\langle binary \rangle$	(383)
		$ \langle binary \rangle \langle bs39 \rangle$	(384)
$\langle bs39 \rangle$	::=	$\langle binary \rangle$	(385)
		$ \langle binary \rangle \langle bs40 \rangle$	(386)
$\langle bs40 \rangle$	::=	$\langle binary \rangle$	(387)
		$ \langle binary \rangle \langle bs41 \rangle$	(388)
$\langle bs41 \rangle$	::=	$\langle binary \rangle$	(389)
		$ \langle binary \rangle \langle bs42 \rangle$	(390)
$\langle bs42 \rangle$	::=	$\langle binary \rangle$	(391)
		$ \langle binary \rangle \langle bs43 \rangle$	(392)
$\langle bs43 \rangle$	::=	$\langle binary \rangle$	(393)
		$ \langle binary \rangle \langle bs44 \rangle$	(394)
$\langle bs44 \rangle$	::=	$\langle binary \rangle$	(395)
		$ \langle binary \rangle \langle bs45 \rangle$	(396)
$\langle bs45 \rangle$	::=	$\langle binary \rangle$	(397)
		$ \langle binary \rangle \langle bs46 \rangle$	(398)
$\langle bs46 \rangle$	::=	$\langle binary \rangle$	(399)
		$ \langle binary \rangle \langle bs47 \rangle$	(400)

$\langle bs47 \rangle$	$::=$	$\langle binary \rangle$	(401)
		$ \langle binary \rangle \langle bs48 \rangle$	(402)
$\langle bs48 \rangle$	$::=$	$\langle binary \rangle$	(403)
		$ \langle binary \rangle \langle bs49 \rangle$	(404)
$\langle bs49 \rangle$	$::=$	$\langle binary \rangle$	(405)
		$ \langle binary \rangle \langle bs50 \rangle$	(406)
$\langle bs50 \rangle$	$::=$	$\langle binary \rangle$	(407)
		$ \langle binary \rangle \langle bs51 \rangle$	(408)
$\langle bs51 \rangle$	$::=$	$\langle binary \rangle$	(409)
		$ \langle binary \rangle \langle bs52 \rangle$	(410)
$\langle bs52 \rangle$	$::=$	$\langle binary \rangle$	(411)
		$ \langle binary \rangle \langle bs53 \rangle$	(412)
$\langle bs53 \rangle$	$::=$	$\langle binary \rangle$	(413)
		$ \langle binary \rangle \langle bs54 \rangle$	(414)
$\langle bs54 \rangle$	$::=$	$\langle binary \rangle$	(415)
		$ \langle binary \rangle \langle bs55 \rangle$	(416)
$\langle bs55 \rangle$	$::=$	$\langle binary \rangle$	(417)
		$ \langle binary \rangle \langle bs56 \rangle$	(418)
$\langle bs56 \rangle$	$::=$	$\langle binary \rangle$	(419)
		$ \langle binary \rangle \langle bs57 \rangle$	(420)
$\langle bs57 \rangle$	$::=$	$\langle binary \rangle$	(421)
		$ \langle binary \rangle \langle bs58 \rangle$	(422)
$\langle bs58 \rangle$	$::=$	$\langle binary \rangle$	(423)
		$ \langle binary \rangle \langle bs59 \rangle$	(424)
$\langle bs59 \rangle$	$::=$	$\langle binary \rangle$	(425)
		$ \langle binary \rangle \langle bs60 \rangle$	(426)
$\langle bs60 \rangle$	$::=$	$\langle binary \rangle$	(427)
		$ \langle binary \rangle \langle bs61 \rangle$	(428)
$\langle bs61 \rangle$	$::=$	$\langle binary \rangle$	(429)
		$ \langle binary \rangle \langle bs62 \rangle$	(430)
$\langle bs62 \rangle$	$::=$	$\langle binary \rangle$	(431)
		$ \langle binary \rangle \langle bs63 \rangle$	(432)
$\langle bs63 \rangle$	$::=$	$\langle binary \rangle$	(433)
			(434)